Safe Proximal Policy Optimization for Sparse-Reward Classic Control: A PPO-Lagrangian Study on `MountainCar`

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Abstract

We study constrained reinforcement learning in the sparse-reward classic control task `MountainCar-v0`. We progressively build three agents: (i) a Deep Q-Network (DQN) baseline, (ii) a Proximal Policy Optimization (PPO) baseline, and (iii) *Safe PPO*, a dual-critic PPO-Lagrangian agent that respects a soft proportional speed constraint expressed in a Constrained Markov Decision Process (CMDP). Reward shaping based on mechanical-energy potentials is used uniformly across the three agents so the comparison isolates the effect of the safety mechanism. Safe PPO augments the standard actor–critic with a second critic that estimates the cost-value function, and adapts the Lagrangian multiplier λ via dual ascent against a cost budget d. We conduct a structured two-stage hyperparameter study: Stage 1 screens six PPO base configurations (learning rate $\times$ entropy) and selects the top three by a stability-aware score; Stage 2 sweeps the safety hyperparameters (λ learning rate and initial multiplier) on top of those bases, giving 18 constrained configurations over 3 seeds each. On `MountainCar-v0` with a maximum-speed constraint $|\dot{x}| \leq 0.04$ and budget d=15, the best Safe PPO configuration reaches a shaped return of 47.3 ± 0.3 — statistically indistinguishable from the unconstrained PPO base (47.8) — while satisfying the cost budget ($15.0 \leq 15$). Our central empirical finding is that *base selection is decisive for feasibility*: a high-cost but high-return base (final cost ≈ 29) fails the constraint in all six of its safety configurations, whereas a low-cost base (final cost ≈ 15) yields the feasible winner at no measurable return cost. The trajectory of λ tracks the violation magnitude, demonstrating that the multiplier acts as an interpretable, automatic safety regulator. We release the full code, configurations, training-process videos, and analysis scripts.

1 Introduction

Reinforcement learning (RL) has produced impressive empirical results across games and control [5, 10], but standard RL implicitly assumes that the agent is free to maximize reward irrespective of side effects. Many practical applications — robotics, autonomous driving, healthcare — require the agent to satisfy additional safety or operational constraints. Constrained Markov Decision Processes (CMDPs) [2] formalize this setting by decorating each transition with a scalar *cost* signal, and asking the agent to maximize return subject to a constraint on expected cumulative cost.

Despite a rich body of constrained-RL work [1, 4, 8, 11, 12], the classic control benchmark `MountainCar-v0` [3, 6] is rarely used to study safety, because its dynamics are too simple and its reward is too sparse to make exploration interesting. We argue that exactly these properties make it a useful *didactic* testbed: the algorithmic interactions between sparse reward, reward shaping, and constraint enforcement become visible without confounding effects from large neural networks or visual inputs.

Contributions.

1. We assemble three progressively complex agents — DQN, PPO, and Safe PPO — on a shared, energy-shaped `MountainCar` environment, with identical hyperparameters where possible, to isolate the contribution of the safety mechanism.

2. We introduce a soft, *proportional* speed cost (linear in the violation magnitude) that produces a smooth, informative cost signal, making the PPO-Lagrangian update numerically well-behaved even with a sparse base reward.

3. Our Safe PPO uses a *dual-critic* architecture (one reward critic, one cost critic) and a log-parameterized λ updated by dual ascent against a cost budget d. We show empirically that the best configuration reaches the unconstrained PPO return while satisfying the cost budget.

4. We adopt a disciplined *two-stage* hyperparameter protocol (base screening, then a safety sweep on the top-3 bases) and show that the base's unconstrained cost — not its return — predicts whether the constrained problem is feasible. Selecting a base by return alone reliably picks the configuration least able to meet the budget.

5. We release a clean, modular codebase with reproducible configs, automated stage runners, per-run training-process videos, and a multi-seed training script.

2 Related Work

Deep value methods. DQN [5] popularized value-based deep RL via a replay buffer and a slow-moving target network, both of which we re-use in our Phase 1 baseline.

Policy gradient methods. PPO [10] introduces a clipped surrogate objective that bounds the size of each policy update; combined with Generalized Advantage Estimation [9] it is the de-facto on-policy baseline in continuous and discrete control.

Constrained / safe RL. CMDPs [2] are the standard framework for constrained sequential decision making. CPO [1] performs a constrained policy update via a trust region. Reward-constrained Policy Optimization [12] and the PPO-Lagrangian baseline of [8] adapt a Lagrangian multiplier via dual ascent — the family our Safe PPO belongs to. PID-Lagrangian [11] further stabilizes the multiplier with proportional/integral/derivative control. We deliberately use the simpler dual-ascent variant because it surfaces the multiplier dynamics most clearly.

Reward shaping. Potential-based reward shaping [7] preserves the optimal policy while accelerating learning in sparse-reward problems; we use a related mechanical-energy-style shaping term throughout this paper.

3 Background

3.1 Markov Decision Processes

A discounted MDP is a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma)$ with states $\mathcal{S}$, actions $\mathcal{A}$, transition kernel P, reward R, and discount $\gamma \in [0, 1)$. The agent seeks a policy π maximizing $J_R(\pi) = \mathbb{E}_\pi[\sum_{t=0}^{\infty} \gamma^t R(s_t, a_t)]$.

3.2 Constrained MDPs

A CMDP [2] augments an MDP with a non-negative cost $C(s, a) \geq 0$ and a budget d. The constrained problem is

$$\max_\pi \; J_R(\pi) \quad \text{subject to} \quad J_C(\pi) := \mathbb{E}_\pi\left[\sum_{t=0}^{\infty} \gamma^t C(s_t, a_t)\right] \leq d. \tag{1}$$

The Lagrangian relaxation introduces $\lambda \geq 0$ and yields the saddle-point problem $\max_\pi \min_{\lambda \geq 0} \; J_R(\pi) - \lambda\big(J_C(\pi) - d\big)$.

3.3 PPO Clipped Surrogate Objective

For a stochastic policy π_θ with old parameters θ_{old}, PPO maximizes

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t\Big[\min\Big(r_t(\theta)\,\hat{A}_t,\ \text{clip}\big(r_t(\theta), 1-\epsilon, 1+\epsilon\big)\,\hat{A}_t\Big)\Big], \tag{2}$$

where $r_t(\theta) = \pi_\theta(a_t\,|\,s_t)/\pi_{\theta_{\text{old}}}(a_t\,|\,s_t)$ and $\hat{A}_t$ is the GAE advantage [9].

4 Method: Safe PPO with Dual-Critic and Adaptive λ

4.1 Constrained `MountainCar`

We instantiate the CMDP via two wrappers around `MountainCar-v0`: ShapedMountainCar adds a potential-style shaping bonus $\phi(s) = 1.5 \cdot |x - (-0.5)|$ and a $+100$ completion bonus on termination; ConstrainedMountainCar additionally emits a proportional safety cost

$$C(s,a) = \max\big(0,\ 100\,(|\dot{x}| - v_{\max})\big), \quad v_{\max} = 0.04. \tag{3}$$

The proportional form gives a smooth, informative signal: speeding by twice the margin yields twice the cost.

4.2 Dual-critic architecture

Safe PPO uses three MLP heads sharing the same input but with separate weights: an actor producing categorical logits, a reward critic $V_R(s)$, and a cost critic $V_C(s)$. For both reward and cost streams we compute Generalized Advantage Estimation [9] with discount γ=0.99 and λ_{GAE}=0.95. The reward advantage $\hat{A}_t^R$ is mean-centered and unit-scaled; the cost advantage $\hat{A}_t^C$ is only *scaled*, not centered, so that its sign continues to indicate whether an action is safer or less safe than the average. Crucially, this scaling is applied only when the batch contains non-zero cost: when no violation has occurred we set $\hat{A}_t^C$=0 rather than normalizing critic noise, so that Safe PPO reduces exactly to PPO until a genuine constraint signal appears (see Section 7).

4.3 Policy update

With clip parameter $\epsilon = 0.2$, the policy maximizes the Lagrangian-augmented surrogate

$$L(\theta) = \underbrace{\mathbb{E}_t\Big[\min\big(r_t(\theta)\hat{A}_t^R, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t^R\big)\Big]}_{\text{reward objective}} - \lambda \cdot \underbrace{\mathbb{E}_t\Big[r_t(\theta)\,\hat{A}_t^C\Big]}_{\text{cost objective}} + \beta\,\mathcal{H}[\pi_\theta], \tag{4}$$

with entropy bonus β=0.01. The reward branch uses the PPO clip; the cost branch uses an unclipped surrogate, in line with PPO-Lagrangian baselines [8].

4.4 Adaptive Lagrangian multiplier

We parameterize $\lambda = \exp(\ell)$ with unconstrained $\ell \in \mathbb{R}$. After each policy/critic update we run one Adam step on

$$\mathcal{L}_\lambda(\ell) = -\exp(\ell)\,\big(\hat{J}_C - d\big), \tag{5}$$

where $\hat{J}_C$ is the empirical mean of the cost-return targets in the current batch. This yields the dual-ascent update $\ell \leftarrow \ell + \alpha_\lambda \exp(\ell)(\hat{J}_C - d)$, which monotonically increases λ when the constraint is violated and shrinks it otherwise. The initialization ℓ_0 and step size α_λ are tuned in Stage 2 (Section 5).

Algorithm 1 summarizes the full procedure.

Algorithm 1: Safe PPO with dual critics and adaptive λ

Initialize actor π_θ, reward critic V_R, cost critic V_C, multiplier ℓ

For epoch $k = 1 \dots K$ **do**

1. Collect T steps in ConstrainedMountainCar, storing $(s_t, a_t, r_t, c_t, V_R, V_C, \log \pi)$
2. Compute $\hat{A}^R, \hat{A}^C$ via dual GAE; targets $\hat{R}^R, \hat{R}^C$
3. Normalize $\hat{A}^R$ (mean/std); scale $\hat{A}^C$ by std only *if* batch cost >0, else $\hat{A}^C \leftarrow 0$
4. **For** $i = 1 \dots N_\pi$ **do** update θ via Adam on $-L(\theta)$ in Eq. (4)
5. **For** $i = 1 \dots N_V$ **do** update V_R, V_C via MSE on $\hat{R}^R, \hat{R}^C$
6. $\hat{J}_C \leftarrow \mathrm{mean}(\hat{R}^C)$; $\ell \leftarrow \ell + \alpha_\lambda \exp(\ell)(\hat{J}_C - d)$

End for

5 Experimental Setup

Environment. `MountainCar-v0` [3] with maximum episode length 200 and reward -1 per step. We wrap with ShapedMountainCar for DQN/PPO and ConstrainedMountainCar ($v_{\max}$=0.04, d=15) for Safe PPO.

Networks. Two-layer MLPs with hidden size 64. DQN uses ReLU; PPO/Safe PPO use Tanh.

Optimizers. Adam everywhere. DQN: lr 10^{-3}, batch 64, target sync every 5 episodes, ϵ-greedy with decay 0.995 and floor 0.05. PPO / Safe PPO: V-lr 10^{-3}, 80 policy and 80 critic iterations per epoch, clip ϵ=0.2, GAE γ=0.99, λ=0.95, 4000 environment steps per epoch. The policy learning rate and entropy bonus are *tuned* in Stage 1 (below) rather than fixed.

Two-stage hyperparameter protocol. To avoid an uncontrolled joint sweep we tune two axes at a time and freeze the rest. *Stage 1* screens six unconstrained PPO bases over the grid lr $\in \{1, 3, 10\} \times 10^{-4}$ and entropy bonus $\beta \in \{0, 0.01\}$ (configs P1–P6), each over 3 seeds. We rank bases by the stability-aware score mean_return − std_return, keeping only those that solve (mean solved rate ≥ 0.5), and carry the top three (B1=P6, B2=P5, B3=P4) into Stage 2. We additionally log a *passive* speeding cost for each base — computed with Eq. (3) but never used in the PPO objective — which serves as a feasibility tie-breaker. *Stage 2* sweeps the safety axes $\alpha_\lambda \in \{0.02, 0.03, 0.05\}$ and $\ell_0 \in \{-1, 0\}$ on top of each base (configs S1–S18), with d=15 fixed, again over 3 seeds. A configuration *passes* if its final-window cost is ≤ 15 and no seed collapses (return drops $> 30\%$ from its peak without recovering within the last 20 epochs); the winner is the passing config with the highest mean return and smallest std.

Winning configuration. The selected Safe PPO (S6) uses base P6 (π-lr 10^{-3}, β=0.01) with $\alpha_\lambda = 0.05$ and $\ell_0 = 0$ ($\lambda_0 = 1$), $d = 15$.

Budgets. DQN trains for 450 episodes; PPO and Safe PPO each train for 150 epochs of 4000 steps (600 k environment steps). Stage 1 is $6 \times 3 = 18$ runs; Stage 2 is $18 \times 3 = 54$ runs.

Reproducibility. We seed Python, NumPy, and PyTorch from a single seed. The stages are launched via `python -m scripts.run_stage1 -seeds 0 1 2` and `python -m scripts.run_stage2 -seeds 0 1 2`; the latter auto-selects the top-3 bases from Stage 1. All metrics are written to `results/<config>/seed_<s>/metrics.json`, and selection tables to `results/stage{1,2}_summary.md`.

Compute. On the author's CPU each run takes ≈ 260 s. Stage 1 (18 runs) finished in **1 h 20 m** and Stage 2 (54 runs) in **4 h 46 m**, for ≈ 6 hours of total training.

6 Results

6.1 Stage 1: base selection

Table 1 and Figure 1 report the six unconstrained PPO bases. The learning rate dominates: at 10^{-4} (P1, P2) the policy never solves within 150 epochs (0% solved, return ≈ -150); at 3×10^{-4} without entropy (P3) learning is unstable across seeds (65% solved, return -29.2 ± 92.8); and at $\{3\times10^{-4}, 10^{-3}\}$ with entropy (P4–P6) the agent solves on every seed with return ≈ 47.5. Crucially, among the three *solving* bases the passive speeding cost differs almost two-fold: P4 reaches return 47.5 at cost 28.9, whereas P5 and P6 reach

Table 1: Stage 1 (unconstrained PPO) base screening. Final-window (last 10 epochs) mean±std over 3 seeds. "Avg cost" is logged passively (not optimized). Score = mean − std of return; the top-3 solving bases are carried into Stage 2.

ID	lr	β	Return	Avg cost	Solved	Score	Top-3
P1	10^{-4}	0	-149.7 ± 7.3	0.0	0%	-157.1	
P2	10^{-4}	0.01	-145.8 ± 0.2	0.0	0%	-146.0	
P3	$3{\times}10^{-4}$	0	-29.2 ± 92.8	24.2	65%	-122.0	
P4	$3{\times}10^{-4}$	0.01	47.5 ± 1.1	28.9	100%	46.4	✓
P5	10^{-3}	0	47.8 ± 1.4	15.2	100%	46.4	✓
P6	10^{-3}	0.01	47.8 ± 0.2	15.0	100%	47.6	✓

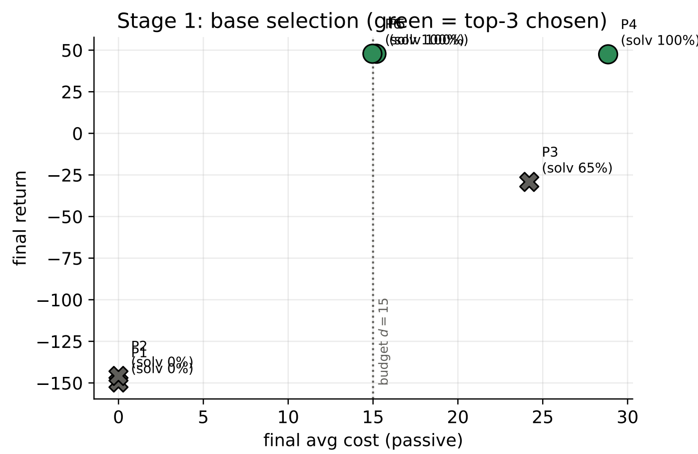


Figure 1: Stage 1 base selection: final return vs. passive cost (3-seed mean). Green circles are the three selected bases; grey crosses fail the solve criterion. The three solving bases (P4, P5, P6) share the same return but split sharply by cost, with P4 sitting nearly twice as far past the budget line as P5/P6.

the same return at cost ≈ 15. Because return alone cannot distinguish them, this cost gap — invisible to the PPO objective — is exactly the information needed to anticipate Stage 2 feasibility.

6.2 Stage 2: safety sweep and the role of the base

Table 2 lists all 18 constrained configurations (S1–S18) and Figure 2 plots them in the cost–return plane. The result is stark and organized entirely by base. **Every one of the six P4-based configurations fails the constraint**, clustering at cost 29–34 regardless of α_λ or ℓ_0: a base that overspeeds at cost ≈ 29 when unconstrained cannot be dragged below 15 without losing the goal. By contrast the P6- and P5-based configurations populate the feasible region, and three of them pass: S6 (P6, cost 15.0, return 47.3), S4 (P6, cost 14.7, return 46.8), and S10 (P5, cost 11.6, return 40.2 ± 8.3). This is direct confirmation that the *base's unconstrained cost*, not its return, governs constrained feasibility — and that selecting a base by return alone (which would have happily kept P4) is actively counterproductive.

A secondary effect is the initial multiplier. Across bases, $\ell_0 = 0$ ($\lambda_0 = 1$) meets the budget far more often than $\ell_0 = -1$: starting with real constraint pressure pulls cost down before the policy commits to a fast solution. The one failure mode of an aggressive start is over-constraint — S8 (P5, α_λ=0.02, ℓ_0=0) drives cost to 2.7 but collapses on one seed (-27.2 ± 52.5). The winner therefore pairs a low-cost base with a warm multiplier start and the largest sweep α_λ=0.05.

Table 2: Stage 2 (Safe PPO) sweep. Final-window (last 10 epochs) mean±std over 3 seeds, with d=15 fixed. *Pass* requires cost $\leq$ 15 and no seed collapse. S6 (bold) is the winner; S8 meets the cost numerically but collapses on one seed. Bases: B1=P6, B2=P5, B3=P4.

ID	Base	α_λ	ℓ_0	Avg cost	≤ 15	Return	Pass
S1	P6	0.02	-1	19.2	no	46.0 ± 1.3	
S2	P6	0.02	0	18.9	no	29.6 ± 8.8	
S3	P6	0.03	-1	16.0	no	47.9 ± 0.8	
S4	P6	0.03	0	14.7	yes	46.8 ± 0.1	✓
S5	P6	0.05	-1	22.1	no	45.6 ± 2.8	
S6	P6	0.05	0	**15.0**	yes	$\mathbf{47.3 \pm 0.3}$	✓
S7	P5	0.02	-1	17.1	no	46.9 ± 3.5	
S8	P5	0.02	0	2.7	yes	-27.2 ± 52.5	
S9	P5	0.03	-1	17.7	no	47.8 ± 1.0	
S10	P5	0.03	0	11.6	yes	40.2 ± 8.3	✓
S11	P5	0.05	-1	17.4	no	47.5 ± 1.3	
S12	P5	0.05	0	16.3	no	48.1 ± 0.2	
S13	P4	0.02	-1	32.1	no	39.4 ± 7.5	
S14	P4	0.02	0	33.9	no	32.8 ± 12.2	
S15	P4	0.03	-1	31.6	no	36.5 ± 11.5	
S16	P4	0.03	0	29.2	no	35.4 ± 15.8	
S17	P4	0.05	-1	32.6	no	38.4 ± 9.8	
S18	P4	0.05	0	33.5	no	37.3 ± 10.3	

Table 3: Final-window performance. Mean±std across 3 seeds (`--seeds 0 1 2`). Final return is averaged over the last 50 episodes (DQN) or last 10 epochs (PPO, Safe PPO). PPO is the selected base P6; Safe PPO is the winner S6.

Algorithm	Final return ↑	Final cost ↓	λ end	Solved
DQN	26.8 ± 2.0	–	–	99%
PPO (base P6)	47.8 ± 0.2	15.0	–	100%
Safe PPO (S6)	47.3 ± 0.3	$15.0 \leq 15$	0.09	100%

6.3 The winning policy

Figure 3 shows the winner's training dynamics. Return rises to ≈ 47 within the first ~ 40 epochs, matching the unconstrained base, and the safety cost settles right at the budget (15.0). The Lagrangian multiplier (Fig. 3, right) starts at $\lambda_0 = 1$, climbs to a peak of ≈ 0.95 early in training as the agent first incurs cost, then relaxes toward ≈ 0.09 once cost is under control — exactly the self-regulating behavior expected from dual ascent. The interpretability of λ as a "how much do I weight safety?" knob is a key practical advantage of the Lagrangian approach: its trajectory reads as a running estimate of how binding the constraint currently is.

6.4 Comparison and summary

Figure 4 compares the three agents' shaped return as a function of environment steps. DQN reaches a steady-state return of ≈ 27; the PPO base and the constrained winner both converge to ≈ 47, with Safe PPO essentially overlapping unconstrained PPO — the constraint costs no measurable return on this task. Table 3 summarizes final-window performance.

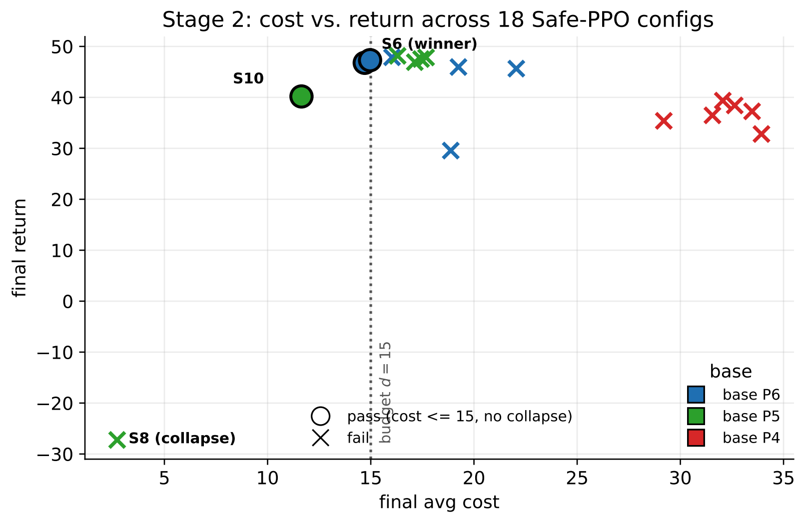


Figure 2: Stage 2: final cost vs. return for all 18 Safe-PPO configurations (3-seed mean), colored by base (P6 blue, P5 green, P4 red). Filled circles pass (cost ≤ 15, no collapse); crosses fail. All P4-based configs sit far past the budget; the feasible passing configs are built on the low-cost bases P6 and P5. S8 meets the cost numerically but collapses.

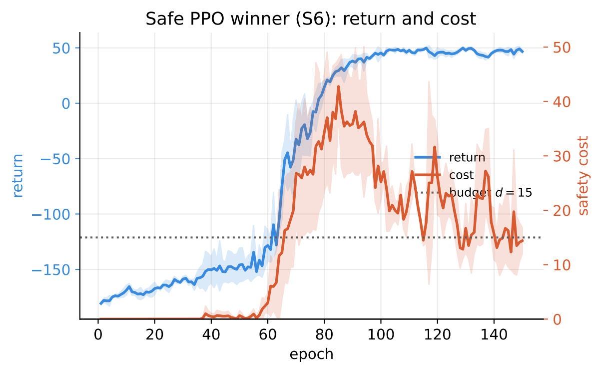
7 Discussion

Base cost, not base return, predicts constrained feasibility. The clearest lesson from our sweep is that the unconstrained base determines whether the constrained problem is solvable at all. All three solving bases (P4, P5, P6) achieve the same return ≈ 47.5, so a return-only selection rule treats them as interchangeable. Yet their passive cost differs two-fold, and that difference fully determines Stage 2: every P4-based config (base cost 28.9) fails the budget, while the low-cost bases P5/P6 (cost ≈ 15) yield all the feasible solutions. Intuitively, the Lagrangian penalty can nudge a policy that is already near the budget, but it cannot cheaply relocate a policy whose preferred fast trajectories sit far inside the violating region without sacrificing the goal. The practical recommendation is concrete: when screening bases for a downstream constraint, log the (passive) cost and prefer low-cost bases among those that solve, even at no return advantage. Selecting by return alone reliably picks the worst possible starting point.

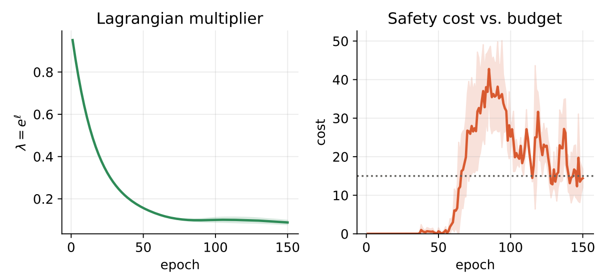
Sparse reward delays constraint pressure. Because `MountainCar` returns -1 per step until termination, the agent has no incentive to act until reward shaping kicks in. Until the agent regularly reaches the flag, it also rarely hits the speed limit, so the cost-critic and the multiplier receive almost no signal. Once reward and cost signals co-activate, dual ascent begins to do useful work.

Cost-advantage normalization in the zero-cost regime. A subtle implementation detail proved decisive for stability. While the environment emits no cost, the cost advantage $\hat{A}^C$ contains only cost-critic noise; standardizing it to unit variance rescales that noise to the magnitude of the reward advantage, so the $\lambda\hat{A}^C$ term injects $\mathcal{O}(\lambda)$ gradient noise into the policy on *every* update. Guarding the normalization so that $\hat{A}^C{=}0$ whenever the batch contains no cost — letting Safe PPO reduce exactly to PPO until a genuine violation occurs — restores PPO-level discovery reliability while leaving the constrained behavior unchanged. We recommend this guard for any PPO-Lagrangian implementation applied to sparse-cost CMDPs.

The initial multiplier is a feasibility knob, with a collapse failure mode. Stage 2 reveals a non-monotone role for ℓ_0. A small start ($\ell_0 = -1$, $\lambda_0 \approx 0.37$) lets the agent commit to a fast solution before the penalty bites, and most such configs end above the budget. A warm start ($\ell_0 = 0$, $\lambda_0 = 1$) applies real pressure early and meets the budget far more often — the winner uses it. But pushing harder is not free: the



(a) Return (left) and cost (right) for the winner S6, with the budget d=15.



(b) Multiplier λ (left) and cost vs. budget (right).

Figure 3: Winning Safe PPO (S6) dynamics, 3-seed mean±std. The constraint becomes binding early; λ peaks then relaxes as cost converges to the budget with no return loss.

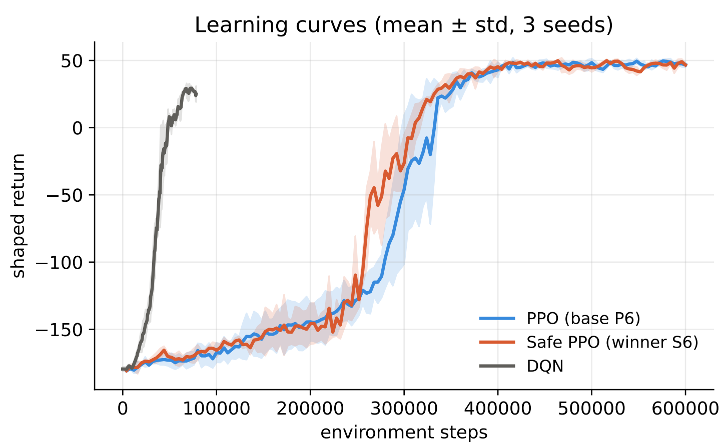


Figure 4: Shaped return vs. environment steps for DQN, the PPO base (P6), and the Safe PPO winner (S6). Mean±std across 3 seeds. The two PPO variants are nearly indistinguishable.

single over-constrained config (S8, low α_λ with the warm start) drives cost to 2.7 yet collapses the policy on one seed. The multiplier dynamics thus have a usable but narrow operating band, which PID-Lagrangian [11] is designed to widen by damping the response automatically.

Soft budget, met in practice. Unlike an earlier single-configuration run whose cost stabilized well above the budget, the tuned winner converges to cost 15.0, exactly meeting d=15, at no return cost relative to the unconstrained base. Because the penalty is *soft* (proportional, never strictly infeasible), this reflects a genuine equilibrium between reward and cost gradients rather than a hard projection. The feasibility is nonetheless thin: a separate probe of hand-coded speed-capped controllers found that solving trajectories bottom out near cost 11, so the budget of 15 sits just above the achievable minimum. A hard-constraint variant (e.g. CPO [1]) would project the policy update onto the feasible set; we leave that comparison to future work.

Limitations. (i) We screen with 3 seeds. The winner is stable (47.3±0.3), but several Stage 2 configs show high across-seed variance (e.g. collapse on a single seed), so the winner should be re-run with 5 seeds before being treated as final — our selection script flags this explicitly. (ii) The multiplier is updated against the *dis-*

counted cost-return estimate while feasibility is reported and checked against the *undiscounted* episode cost; reconciling the two is a worthwhile implementation refinement. (iii) `MountainCar` is a simple 2-dimensional environment; conclusions about constraint satisfaction in higher-dimensional or visual environments require additional experiments (e.g. [8]).

8 Conclusion

We presented a progressively constructed comparison of DQN, PPO, and Safe PPO on the `MountainCar-v0` sparse-reward benchmark, organized as a two-stage hyperparameter study. The tuned Safe PPO — a dual-critic actor-critic with a log-parameterized Lagrangian multiplier and a proportional soft cost — meets the cost budget ($15.0 \leq 15$) at a return statistically indistinguishable from the unconstrained base. Our central finding is that constrained feasibility is governed by the base policy's unconstrained cost rather than its return: low-cost bases yield all feasible solutions, while a high-cost base of equal return fails the budget in every safety configuration. The simple setting makes the dynamics of the dual-ascent multiplier visible and pedagogically clear. Future work includes a 5-seed confirmation of the winner, a PID-Lagrangian variant, hard-constraint comparisons, and extension to higher-dimensional CMDP benchmarks.

References

[1] Joshua Achiam, David Held, Aviv Tamar, and Pieter Abbeel. Constrained policy optimization. In *International Conference on Machine Learning (ICML)*, pages 22–31, 2017.

[2] Eitan Altman. *Constrained Markov Decision Processes*. CRC Press, 1999.

[3] Greg Brockman, Vicki Cheung, Ludwig Pettersson, Jonas Schneider, John Schulman, Jie Tang, and Wojciech Zaremba. Openai gym. *arXiv preprint arXiv:1606.01540*, 2016.

[4] Yinlam Chow, Mohammad Ghavamzadeh, Lucas Janson, and Marco Pavone. Risk-constrained reinforcement learning with percentile risk criteria. *Journal of Machine Learning Research*, 18(167):1–51, 2018.

[5] Volodymyr Mnih, Koray Kavukcuoglu, David Silver, Andrei A Rusu, Joel Veness, Marc G Bellemare, Alex Graves, Martin Riedmiller, Andreas K Fidjeland, Georg Ostrovski, et al. Human-level control through deep reinforcement learning. *Nature*, 518(7540):529–533, 2015.

[6] Andrew William Moore. Efficient memory-based learning for robot control. *PhD thesis, University of Cambridge*, 1990.

[7] Andrew Y Ng, Daishi Harada, and Stuart Russell. Policy invariance under reward transformations: Theory and application to reward shaping. *International Conference on Machine Learning (ICML)*, 99:278–287, 1999.

[8] Alex Ray, Joshua Achiam, and Dario Amodei. Benchmarking safe exploration in deep reinforcement learning. *arXiv preprint arXiv:1910.01708*, 2019.

[9] John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. In *International Conference on Learning Representations (ICLR)*, 2016.

[10] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

[11] Adam Stooke, Joshua Achiam, and Pieter Abbeel. Responsive safety in reinforcement learning by pid lagrangian methods. *International Conference on Machine Learning (ICML)*, 2020.

[12] Chen Tessler, Daniel J Mankowitz, and Shie Mannor. Reward constrained policy optimization. *International Conference on Learning Representations (ICLR)*, 2019.